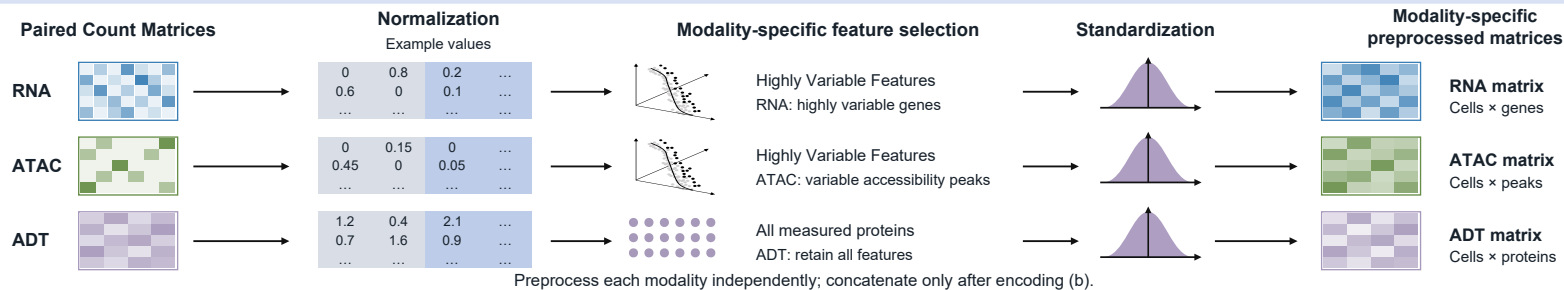


a Input Data & Processing

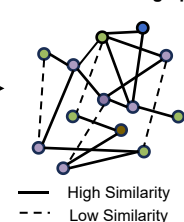


c Graph-Aware Residual Quantization (GARQ)

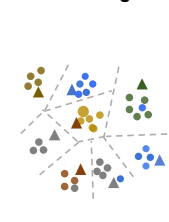
Latent Embeddings



Batch-local KNN graph



Anchor Assignment



Cell-anchor assignments use both cosine similarity and neighbors within the current mini-batch.

n : cells in current mini-batch; $\alpha = 0.2$

$$Q(h_i) = a_k^*, k^*$$

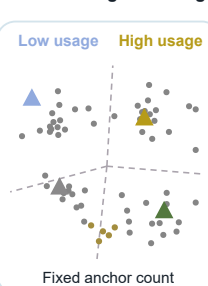
$$= \arg \max_{k=1, \dots, K} \left\{ \cos(h_i, a_k) + \alpha \sum_{j=1}^n s_{ij} \cos(h_j, a_k) \right\}$$

Rare-State Preservation

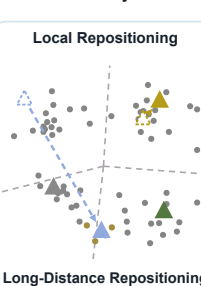


d Dynamic Anchor Adjustment (GARQ)

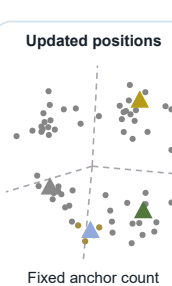
1. EMA Usage Tracking



2. Anchor Adjustment



3. Anchor Balance



Continuous usage-weighted updates

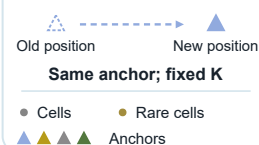
$$\text{EMA: } U_k^t = \eta U_k^{t-1} + (1 - \eta) n_k^t / n^t$$

Long-distance: every quantized mini-batch

Local: activated when $\sum_k U_k + 10^{-4} \geq 1$

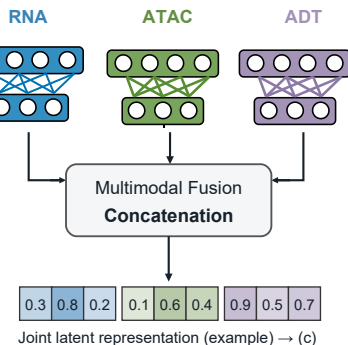
Anchor identities and total K are preserved.

Anchor repositioning



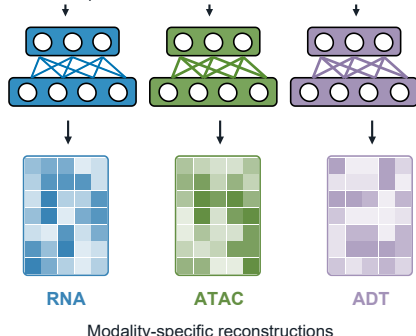
b Multimodal Encoding

Modality-specific cell-wise encoders



e Multimodal Decoding

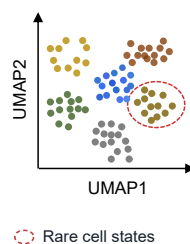
Continuous (b) + quantized (c-d) embeddings
Separate cell-wise reconstruction branches



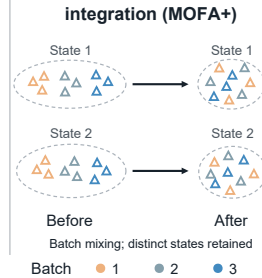
f Downstream Applications (outside GARQ)

Input: GARQ-derived metacell profiles
Per-anchor mean of member-cell profiles

Visualization



Downstream batch-aware integration (MOFA+)



Biological Analysis

